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14. Fabricating Systems A production plant has two fabricating systems, both of which are maintained at 2-week intervals. However, one system is twice as old as the other. The number of finished products fabricated daily by each of the systems is recorded for 30 working days, with the results given in the table. Do these data present sufficient evidence to conclude that the variability in daily production warrants increased maintenance of the older fabricating system? Use the p -value approach.

New System	Old System
$\bar{x}_1 = 246$	$\bar{x}_2 = 240$
$s_1 = 15.6$	$s_2 = 28.2$

$$H_0 = \frac{\sigma_2^2}{\sigma_1^2} = 1, H_a: \frac{\sigma_2^2}{\sigma_1^2} > 1$$

$$\alpha = 0.05$$

$$F_{STAT} = \frac{s_2^2}{s_1^2} \sim F_{n-1, n-1}$$

$$F^* = 3.2678$$

$$P(F_{29, 29} > 3.2678) < 0.05 < 0.05$$

$$p\text{-value} < 0.05 \Rightarrow \text{Reject } H_0$$